CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

# Chapter 2: Introduction to Linear Polynomials

Topic-wise short notes with solved examples

A linear polynomial has the form $ax + b$. This chapter shows how such expressions describe patterns that grow or fall by a fixed amount, and how they appear as straight lines on a graph.

## 1 Algebraic expressions

- An **algebraic expression** combines numbers, variables and operation signs, for example $4x + 5y + 3$.
- The parts separated by $+$ or $-$ are the **terms**: here $4x$, $5y$ and $3$.
- The letters are the **variables** (earlier called letter-numbers): here $x$ and $y$.
- The number multiplying a variable is its **coefficient**: 4 is the coefficient of $x$ and 5 is the coefficient of $y$.
- A term with no variable is a **constant**: here 3.

**Example 1.1** A rectangular plot is $l$ metres long and $w$ metres wide. Fencing costs ₹90 per metre and levelling costs ₹40 per square metre. Write an expression for the total cost.

SOLUTION

Perimeter $= 2l + 2w$, so fencing costs $90(2l + 2w) = 180l + 180w$.

Area $= lw$, so levelling costs $40lw$.

**Total cost** $= ₹(180l + 180w + 40lw)$

## 2 Polynomials in one variable

- An expression in **one variable** whose powers are whole numbers is called a **polynomial** (a univariate polynomial).
- The **degree** is the highest power of the variable.

| Polynomial | Degree | Name |
|---|---|---|
| $8$ | 0 | Constant polynomial |
| $3z + 7$ | 1 | Linear polynomial |
| $x^2 + 5x + 1$ | 2 | Quadratic polynomial |
| $5y^3 + y^2 + 2y - 1$ | 3 | Cubic polynomial |

- A constant such as 8 has degree 0 because it can be written as $8x^0$.
- A **linear polynomial** has the form $ax + b$ with $a \neq 0$.

**Example 2.1** For the polynomial $x^4 - 3x^3 + 6x^2 - 2x + 7$, write the degree, the coefficient of $x^3$ and the constant term.

SOLUTION

The highest power of $x$ is 4.

**Degree 4; coefficient of $x^3$ is $-3$; constant term is 7**

## 3 Value of a polynomial and linear equations

- To find the **value** of a polynomial, substitute a number for the variable. The polynomial works like a machine: the number is the input and the value is the output.
- A polynomial seen in this way is a **function** of its variable. $2x + 3$ is a linear function and $10x - x^2$ is a quadratic function.
- When a linear polynomial is set equal to a number we get a **linear equation**, which can be solved for the variable.

**Example 3.1** Find the value of $5x - 3$ at $x = -1$ and of $7s^2 - 4s + 6$ at $s = 4$.

SOLUTION

$5(-1) - 3 = -8$

$7(16) - 4(4) + 6 = 112 - 16 + 6 = 102$

**$-8$ and $102$**

**Example 3.2** Two numbers add up to 90, and one is 14 more than the other. Find them.

SOLUTION

Let the smaller number be $x$. Then the other is $x + 14$.

$x + (x + 14) = 90$, so $2x = 76$ and $x = 38$.

**The numbers are 38 and 52.**

**Example 3.3** A rectangle is 3 cm longer than twice its width, and its perimeter is 36 cm. Find its length and width.

SOLUTION

Let the width be $w$ cm. Then the length is $(2w + 3)$ cm.

$2(w + 2w + 3) = 36$, so $6w + 6 = 36$ and $w = 5$.

**Width 5 cm; length 13 cm**

## 4 Linear patterns

- A **linear pattern** is a sequence of numbers in which the difference between consecutive terms is always the same.
- Its $n$th term is a linear expression in $n$: (common difference) $\times\, n +$ (a constant).

- To find the rule, take the common difference as the coefficient of $n$, then adjust the constant so that $n = 1$ gives the first term.

**Example 4.1** Triangles are made in a row with matchsticks, each new triangle sharing a side with the one before. The stages use 3, 5, 7, 9, ... matchsticks. Find the rule, the number needed at stage 15, and the stage that uses 61 matchsticks.

SOLUTION

The difference is 2, so the rule is $2n + c$. At $n = 1$: $2 + c = 3$, so $c = 1$.

Stage 15: $2(15) + 1 = 31$

$2n + 1 = 61$ gives $n = 30$.

**Rule $2n + 1$; 31 matchsticks; stage 30**

**Example 4.2** Can 200 matchsticks make exactly one stage of this pattern?

SOLUTION

$2n + 1 = 200$ gives $n = 99.5$, which is not a whole number.

**No. Every stage uses an odd number of matchsticks.**

## 5 Linear growth and linear decay

- **Linear growth**: a quantity **increases** by the same amount over equal intervals.
- **Linear decay**: a quantity **decreases** by the same amount over equal intervals.
- Both are described by (starting value) + (change per interval) × (number of intervals). The change is positive for growth and negative for decay.

**Example 5.1** A library charges a joining fee of ₹150 and ₹20 for each book borrowed. Write the cost of borrowing $n$ books. How many books were borrowed if the cost is ₹430?

SOLUTION

Cost $= 150 + 20n$. This is linear growth: each extra book adds ₹20.

$150 + 20n = 430$ gives $n = 14$.

**Cost $= ₹(150 + 20n)$; 14 books**

**Example 5.2** A candle 24 cm tall burns down by 1.5 cm every hour. Write its height after $t$ hours. What is its height after 6 hours, and when does it burn out?

SOLUTION

Height $h = 24 - 1.5t$. This is linear decay.

After 6 hours: $24 - 9 = 15$ cm. Burnt out when $24 - 1.5t = 0$, so $t = 16$.

**$h = 24 - 1.5t$; 15 cm; after 16 hours**

## 6 Linear relationships

- Two quantities $x$ and $y$ have a **linear relationship** if $y = ax + b$ for fixed numbers $a$ and $b$.
- $a$ is the change in $y$ for each increase of 1 in $x$. $b$ is the value of $y$ when $x = 0$.
- To find $a$ and $b$ from two known pairs of values, substitute each pair into $y = ax + b$ and solve the two equations.

**Example 6.1** A taxi charges a fixed amount plus a rate per kilometre. A 5 km ride costs ₹110 and a 12 km ride costs ₹215. Find the relation $y = ax + b$ between the fare $y$ and the distance $x$.

SOLUTION

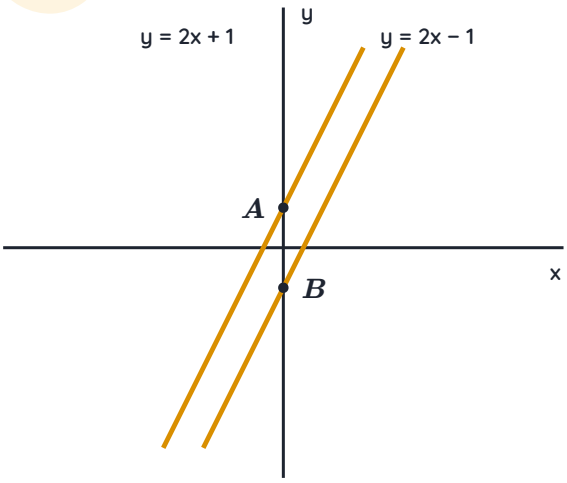
$110 = 5a + b$ and $215 = 12a + b$

Subtracting: $105 = 7a$, so $a = 15$. Then $b = 110 - 75 = 35$.

$y = 15x + 35$: **a fixed charge of ₹35 and ₹15 per km**

## 7 Graph of a linear relationship

- The graph of $y = ax + b$ is a **straight line**. Two points are enough to draw it.
- A point lies on the line only if its coordinates satisfy the equation.
- $a$ is the **slope**: it tells how steep the line is. $b$ is the **$y$-intercept**: the line cuts the $y$-axis at $(0, b)$.
- $y = ax$ always passes through the origin. For $a > 1$ it is steeper than $y = x$, and for $0 < a < 1$ it is less steep.
- A **positive** slope means the line rises to the right (linear growth). A **negative** slope means it falls to the right (linear decay).
- Lines with the **same slope** and different $y$-intercepts are **parallel**.




Both lines have slope 2, so they are parallel. A is (0, 1) and B is (0, −1).

**Example 7.1** Write the slope and the $y$-intercept of $3y = 6x - 12$, and the point where the line cuts the $y$-axis.

SOLUTION

Divide by 3 to get the form $y = ax + b$: $y = 2x - 4$.

**Slope 2; $y$-intercept $-4$; the line cuts the $y$-axis at $(0, -4)$**

**Example 7.2** The graph of a linear polynomial $p(x)$ passes through $(1, 4)$ and $(4, 13)$. Find $p(x)$ and the points where its graph cuts the axes.

SOLUTION

Let $p(x) = ax + b$. Then $a + b = 4$ and $4a + b = 13$.

Subtracting: $3a = 9$, so $a = 3$ and $b = 1$.

It cuts the $y$-axis at $(0, 1)$. It cuts the $x$-axis where $3x + 1 = 0$, that is $x = -\frac{1}{3}$.

**$p(x) = 3x + 1$; the graph cuts the axes at $(0, 1)$ and $\left(-\frac{1}{3}, 0\right)$**

## 8 Common mistakes

- Calling $10x - x^2$ linear because it has an $x$ term. The degree is the **highest** power, here 2.
- Reading the slope from $2y = 4x + 7$ as 4. First divide by 2: $y = 2x + 3.5$, so the slope is 2.
- Forgetting the sign of the coefficient: in $x^4 - 3x^3$ the coefficient of $x^3$ is $-3$.
- Writing the $n$th term of 3, 5, 7, … as $2n$. Check with $n = 1$: the rule must give 3, so it is $2n + 1$.
- Thinking a larger $y$-intercept makes a line steeper. Steepness depends only on the slope.

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